Reliability and Validity

AYHS 20203

Independent Study for November 4

We will not meet in person on November 4. Work through these slides on your own.

- Read the explanations, then try each question before moving to its answer.
- Download the Excel practice workbook. Complete **Practice**, then compare with **Answer Key**.
- Use the final checklist to review your group's Prep 3 work.

Allow about **60–80 minutes**, including practice and your Prep 3 review.

Learning Objectives

By the end of this lesson, you should be able to:

- Distinguish reliability from validity using a measurement example.
- Identify five types of reliability.
- Distinguish measurement validity, internal validity, and external validity.
- Reconstruct a 2×2 table, choose denominators, and interpret association measures.
- Check that your Prep 3 materials answer the assigned descriptive questions.

A Scale That Always Reads High

A wellness team checks a scale with a **known 60 kg calibration mass**.

Reading 1	Reading 2	Reading 3
65 kg	65 kg	65 kg

The readings agree with one another. Each is **5 kg above the reference**.

Would repeating this measurement many more times solve the problem?

Consistency and Accuracy

Reliability concerns consistency under comparable conditions when what we measure has not changed.

Measurement validity concerns whether a measure supports the intended interpretation: does it measure what we want it to measure?

The scale is repeatable but inaccurate. Repeating it does not remove the 5 kg bias. It needs calibration or correction before use.

Reliable measurements can still be wrong.

Why Measurements Differ

Explanation	Example	What to Check
True change	The mass on the scale actually changes	Whether the object changed
Systematic error	The scale reads 5 kg high each time	Calibration against a reference
Random error	Readings fluctuate under otherwise identical conditions	Repeatability and procedures

A change in a score does not, by itself, show that a program worked.

Reliability Over Time

Test–retest reliability: the same people take the same instrument on separate occasions, during a period when the underlying characteristic should be stable.

Intra-rater reliability: the same rater scores the same material again. For example, one instructor scores the same recorded presentation twice.

The key distinction is whether we are studying the **instrument across occasions** or the **repeatability of one person's judgments**.

Reliability Across Raters and Forms

Inter-rater reliability: different raters independently score the same material using the same criteria.

Example: two instructors score the same recorded presentation.

Parallel-forms reliability: people complete two versions designed to measure the same construct with equivalent content and difficulty.

Example: two equivalent forms of a vocabulary test. Merely using two different tests does not make them equivalent.

Internal Consistency

Internal consistency asks whether items intended to measure the **same construct** are related to one another.

A stress scale might ask about feeling overwhelmed and difficulty relaxing. These items should have a meaningful relationship if they reflect the same construct.

Internal consistency alone does not establish that the scale measures stress accurately or covers all important aspects of stress.

Question: Types of Reliability

Match each situation to a reliability type.

1. Two observers independently score the same video.
2. One observer scores that video again a week later.
3. Students repeat the same stable-attribute questionnaire after a week.
4. Students complete equivalent versions A and B of a test.
5. Researchers examine relationships among items on a stress scale.

Choices: **inter-rater, intra-rater, test–retest, parallel-forms, internal consistency**.

Answers: Types of Reliability

Situation	Reliability Type
Two observers	Inter-rater
One observer scoring again	Intra-rater
Same questionnaire over time	Test–retest
Equivalent test versions	Parallel-forms
Related items within a scale	Internal consistency

For repeated measurements, consider actual change, memory, and practice effects before interpreting differences as measurement error.

Improving Measurement Consistency

- Use clear instructions and the same measurement conditions.
- Train raters with a shared scoring rubric and practice examples.
- Check equipment against known references.
- Review ambiguous items and scoring rules.

A reliability coefficient summarizes a particular kind of consistency. Its interpretation depends on the method, population, and intended use. There is no single cutoff that makes every measure acceptable.

Question: A Consistent Error

A scale gives almost identical readings each time, but all readings are 5 kg too high. Which statement is best supported?

A. It is reliable and accurate for measuring mass.

B. It is repeatable, but the readings have a systematic error.

C. It is accurate because repeated readings agree.

D. More repetitions will necessarily remove the error.

Answer: A Consistent Error

B. The readings are repeatable but systematically too high.

Reliability is necessary, but not sufficient, for valid measurement. A measure that changes unpredictably cannot support dependable interpretations.

Consistency alone does not show that the measure captures the intended quantity or construct. One correct reading by chance does not establish validity either.

Three Questions About Validity

Concept	Main Question
Measurement validity	Does the measure capture the intended quantity or construct?
Internal validity	Is the study's conclusion credible, given bias and confounding?
External validity	Do the findings apply to other people, settings, or times?

For a causal claim, internal validity includes whether alternative explanations have been adequately addressed.

Evidence for Measurement Validity

Content validity: does the instrument cover the relevant parts of the construct?

Criterion validity: do scores agree with an appropriate reference or predict a relevant criterion? The quality of the reference matters.

Construct validity: do scores behave as theory predicts, including relationships with related and different constructs?

Validity concerns a particular interpretation and use of scores. Evidence from one population may not establish validity for another.

Example: A Student Stress Questionnaire

Evidence	Main Validity Concept
Experts check coverage of relevant sources of stress	Content validity
Scores agree with an appropriate reference assessment	Criterion validity
Scores relate to overwhelm as theory predicts	Construct validity

Stress has no simple known reference like a calibration mass. One correlation with another questionnaire is only part of the evidence.

Generalizability

A study recruits volunteers from one university's competitive athletics program.

Its measurements could be consistent and accurate, yet its findings may not apply to all college students or to older adults.

Consider who participated, who did not, the setting, and the conditions of measurement.

A larger sample from the same narrow group does not automatically make the findings broadly generalizable.

Question: Which Validity Concern?

A researcher asks whether a finding from university athletes applies to adults living in rural communities. Which concern is this?

A. Internal consistency

B. Intra-rater reliability

C. External validity

D. Parallel-forms reliability

Answer: External Validity

C. External validity concerns generalizability.

The question asks whether findings apply beyond the study's participants and setting.

Good measurement helps, but it does not resolve differences in age, health, activity, or environment between these populations.

Excel Practice A: Calibration Readings

Open **Practice** in the workbook.

Six fictional calibration masses have known reference values and two scale readings each (**A1:D7**).

1. Calculate differences between readings and between each reading and its reference in **G6:I11**.
2. Average each difference column in **G12:I12**.
3. Calculate the correlation between the reference and Reading 1 in **G13**.
4. Write your interpretation before opening **Answer Key**.

Excel Formulas for Practice A

Cell	Formula	Meaning
G6	`=D2-C2`	Reading 2 minus Reading 1
H6	`=C2-B2`	Reading 1 minus reference
I6	`=D2-B2`	Reading 2 minus reference
G12	`=AVERAGE(G6:G11)`	Mean difference
G13	`=CORREL(B2:B7,C2:C7)`	Reference–reading correlation

Fill G6:I6 down through row 11. Copy G12 across through I12. Keep the original data in A2:D7 unchanged.

Practice A: What the Results Mean

- Mean difference between repeated readings: **0 kg**.
- Mean difference from the reference, for either reading: **+5 kg**.
- Correlation between reference and Reading 1: **r = 1.000**.

The repeated readings agree perfectly, but the scale is consistently 5 kg high.

Correlation measures linear association. It does not establish agreement or accuracy. The paired differences reveal the systematic error.

Categorical Refresher: A One-Week Cohort

Now use a separate set of **20 fictional students**. All begin without fatigue and complete one week of follow-up.

- **Short sleep:** usual sleep below 7 hours per night.
- **Not short:** usual sleep of at least 7 hours per night.
- **Outcome:** a reported new fatigue episode during follow-up.

This teaching example uses a defined follow-up period so we can calculate risks. It makes no claim about real students or a clinical sleep threshold.

Excel Practice B: A 2×2 Table

Use rows **13–32** on **Practice**. Complete the table in **G23:I25**.

Sleep Group	New Fatigue: Yes	New Fatigue: No	Total
Short	?	?	?
Not short	?	?	?
Total	?	?	20

Count each student once. The four interior cells should add to 20.

If Excel is unavailable, tally the categories by hand and use a calculator.

Excel Counts and Totals

In **G23**, enter:

```
=COUNTIFS($B$13:$B$32,$F23,$C$13:$C$32,G$22)
```

This counts rows where the sleep group matches F23 and the outcome matches G22.

- Copy across to H23, then down through row 24.
- In I23, enter `=SUM(G23:H23)` and copy to I24.
- In G25, enter `=SUM(G23:G24)` and copy across to I25.

The `$` signs keep the data ranges and appropriate labels fixed as you copy.

Question: Choosing a Denominator

Sleep Group	New Fatigue: Yes	New Fatigue: No	Total
Short	6	4	10
Not short	2	8	10
Total	8	12	20

Before calculating: which denominator answers “What proportion of short sleepers developed fatigue?”

A. 6 B. 8 C. 10 D. 20

The Denominator Defines the Question

C. Use the 10 short sleepers as the denominator.

Question	Calculation	Result
Risk among short sleepers	6 / 10	60%
Risk among students without short sleep	2 / 10	20%
Short sleepers among those with fatigue	6 / 8	75%

The last row describes the exposure distribution **among cases**. It is not the risk among short sleepers.

Risk Ratio and Risk Difference

Complete **G27:G30** in the workbook.

Measure	Excel Formula	Answer
Short-sleep risk	`=G23/I23`	60%
Other-group risk	`=G24/I24`	20%
Risk ratio	`=G27/G28`	3.00
Risk difference	`=G27−G28`	40 percentage points

The one-week risk was **3 times as high**, or **40 percentage points higher**, among short sleepers in this fictional sample.

Odds Ratio Refresher

Odds compare those with the outcome with those without it.

- Short-sleep odds: **6 / 4 = 1.5**.
- Other-group odds: **2 / 8 = 0.25**.
- Odds ratio: **1.5 / 0.25 = 6.00**.

In **G31**, use `=(G23/H23)/(G24/H24)`.

Here, **OR = 6** and **RR = 3**. Report the measure you calculated. “Six times the odds” does not mean “six times the risk.”

Interpretation and Measurement Limits

The sample shows an association between short sleep and subsequent fatigue. It does not establish causation.

- **Measurement:** did everyone interpret “fatigue” in the same way?
- **Internal validity:** could other differences between the groups explain the association?
- **External validity:** would this volunteer sample represent other students or adults?

This exercise is descriptive. We will address inference later. Do not interpret these calculations as a test of statistical significance.

Prep 3 Application

Return to your group's assigned project dataset and variables.

1. Check the required continuous-variable summaries.
2. Inspect the histogram's labels and description of shape and unusual observations.
3. Inspect the scatterplot's axes, pairing, and description of direction, strength, and form.
4. Include the required interpretations and individual contributions.

Today's categorical calculations are lesson practice. They add no graded deliverable to Prep 3.

Prep 3 Procedures and Deadline

- Use the **Continuous Data Summaries and Graphs** lesson for summary and histogram procedures.
- Use the **Scatterplots and Correlation** lesson for the Excel scatterplot procedure.
- Follow your assigned variables and the current Prep 3 instructions in D2L. Prep 3 remains descriptive; r and r^2 are not added requirements.

Prep 3 is due November 4 at 11:30 p.m. Central. The canceled class meeting does not change that deadline.

Final Check Before Quiz 4

Without looking back, explain:

- How a measurement can be reliable but invalid.
- How inter-rater, intra-rater, and test–retest reliability differ.
- What internal consistency and parallel-forms reliability assess.
- How measurement validity differs from internal and external validity.

Also review the earlier lessons on descriptive statistics, z-scores, scatterplots, correlation matrices, r^2, and association versus causation for **Quiz 4 on November 9**.

Sources and Further Reading

- Course lecture materials: *Correlations and Reliability/Validity*, reliability and validity section. Huck, Chapter 4, is optional supporting reading.
- COSMIN definitions of measurement properties, Table 1.1. Supports the reliability and measurement-validity distinctions.
- ETS: Test Reliability, Basic Concepts. Additional explanations of reliability types.
- CDC Field Epidemiology Manual: Analyzing and Interpreting Data. Reference for 2×2 tables and association measures.

All calibration and cohort observations in this lesson and its workbook are fictional teaching data.